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A Bioinspired Defect-Tolerant Hydrogel Medical Patch for Abdominal Wall Defect Repair

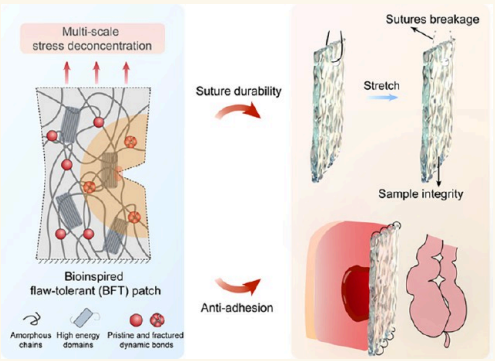
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ABSTRACT: **Point-wise suturing is the standard method for ensuring that patches effectively perform mechanically supportive functions in tissue repair. However, stress concentrations around suture holes can compromise the mechanical stability of patches. In this study, we develop a suturable hydrogel patch with flaw-tolerance capabilities by leveraging multiscale stress deconcentration, inspired by natural silk. This design mitigates stress concentration across two scales through the synergistic integration of nanoscale high-energy crystalline domains and intermolecular interactions. The resulting integral hydrogel patch exhibits superior flaw resistance compared to conventional patches and effectively addresses tissue adhesion issues. To validate the efficacy of the patch, we demonstrate successful *in vivo* repair of abdominal wall defects in rats, comparing the performance of the proposed patch to commercial mesh patches (Prolene). The integral patch design strategy present here offers a valuable approach for developing patches that can be tailored to meet the mechanical support needs of various tissue repair applications.**

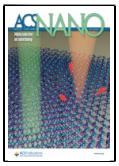


KEYWORDS: *hydrogel patch, biomimetic structure, stress deconcentration, suture resistance, tissue repair*

Biomedical patches serve as versatile therapeutic platforms with applications ranging from ultrasound monitoring[1,2] to conductive cardiac support.[3,4] Clinically, patches are primarily used for mechanical support in areas such as the dura, abdominal wall, rotator cuff, bone regeneration, and vasculature.[5,6] The mechanical robustness of these patches is essential for long-term stability.[7] However, the human body constant motion—comprising both single large-range movements (e.g., exercise)[8,9] and cyclic small-range motions (e.g., breathing, heartbeat)[10,11]—imposes significant and varying stresses that increase the risk of patch fatigue and failure. Suturing, the standard method for securing patches, introduces flaws that cause stress concentrations, heightening the likelihood of mechanical degradation (Figure 1A). These combined mechanical demands and suturing flaws can lead to postoperative complications,[12−15] highlighting an urgent need for improved strategies to enhance patch durability.

To avoid stress concentration, recent advances in patch fixation methods, such as bioadhesion[16−18] and microneedle techniques, show promise.[19] However, the ability of these minimally invasive approaches to withstand the high, instantaneous stresses from large-range movements or the repetitive stresses of cyclic small-range motions remains to be well investigated. Alongside improving fixation methods, enhancing the intrinsic flaw tolerance of the patch itself is a practical strategy for boosting durability. Transforming patches from integral to woven structures can prevent mechanical failure by reducing stress concentration,[20] but while mesh forms are beneficial in surgery, they often cause tissue adhesion[21−23] and are unsuitable for flexible electronics requiring integral structures.[24] Nature offers inspiration for flaw-tolerant materials; spider silk, for instance, maintains remarkable mechanical resilience despite inherent flaws.[25] Even with defects, spider silk withstands substantial and repetitive stresses without easily breaking. This resilience is attributed to multiscale stress deconcentration through non-covalent interactions.[26] Nanocrystalline β-sheets deform and unfold under stress, while weaker hydrogen bonds among

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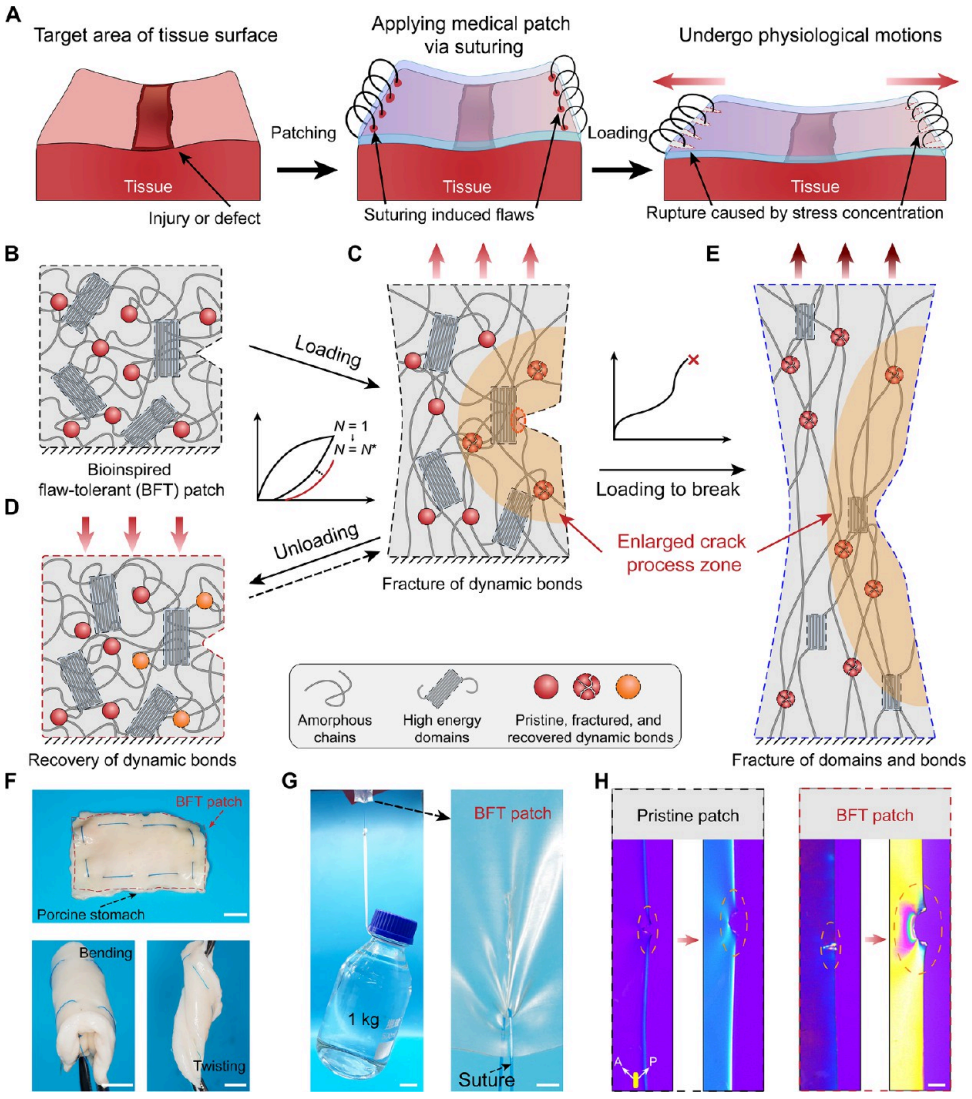


Figure 1. Design and fabrication of BFT patch. (A) Schematic illustration of the typical application of a medical patch via suturing, showing rupture due to induced flaws and stress concentration. (B-E) Mechanism of stress deconcentration in BFT patches: (B) The initial state of the BFT patch with high-energy nanocrystalline domains and dynamic hydrogen bonds. (C) Under loading, dynamic bonds at the crack tip fracture, leading to an enlarged crack process zone. (D) During unloading, dynamic bonds recover, partially restoring the integrity. (E) Under large-range loading to break, both high-energy domains and dynamic bonds undergo fracture, effectively deconcentrating stress. (F) BFT patch being sutured to a porcine stomach, demonstrating flexibility under bending and twisting. Scale bar: 5 mm. (G) Demonstration of the robustness of BFT patch, supporting a 1 kg weight under sutured conditions. Scale bars: 2 cm for left, 1 mm for right. (H) Polarized light images comparing the pristine and BFT patches under loading, showing improved stress distribution and crack blunting in the BFT patch. Scale bar: 1 mm.

amorphous chains undergo a stick−slip mode, breaking and reforming to restore mechanical integrity.[27,28]

Inspired by spider silk multiscale stress-deconcentration strategy, here we designed a flaw-tolerant patch incorporating noncovalent interactions across two scales (Figure 1B). Under load, stress at the crack tip disperses first through the soft amorphous phase, then into rigid nanoscale crystalline domains. The crystalline domains transfer stress to the surrounding amorphous matrix, creating a broad stress distribution zone. Strong intermolecular interactions in the amorphous phase repeatedly break and reform, dissipating energy and blunting crack tips (Figure 1C). Under small strains, reversible interactions in the amorphous phase restore dissipation, while the crystalline domains prevent crack propagation (Figure 1D). At high strain, crystalline domains unfold and align, absorbing destructive energy and preventing rapid crack growth (Figure 1E). Following this multiscale stress-deconcentration principle, we developed a bioinspired flaw-tolerant (BFT) patch and evaluated its mechanical performance under large deformations and cyclic loading, even with significant defects. The BFT patch demonstrated flaw tolerance, mechanical robustness, and biocompatibility comparable to commercial polypropylene mesh (Prolene) and dramatically reduced tissue adhesion in abdominal wall repair. This innovation highlights the potential of the BFT patch to improve soft tissue repair durability and mitigate complications associated with traditional mesh-based solutions.

RESULTS AND DISCUSSION

Design and Fabrication of BFT Patch. To construct the BFT patch, we chose poly(vinyl alcohol) (PVA), a biocompatible polymer with hydroxyl-rich side chains that

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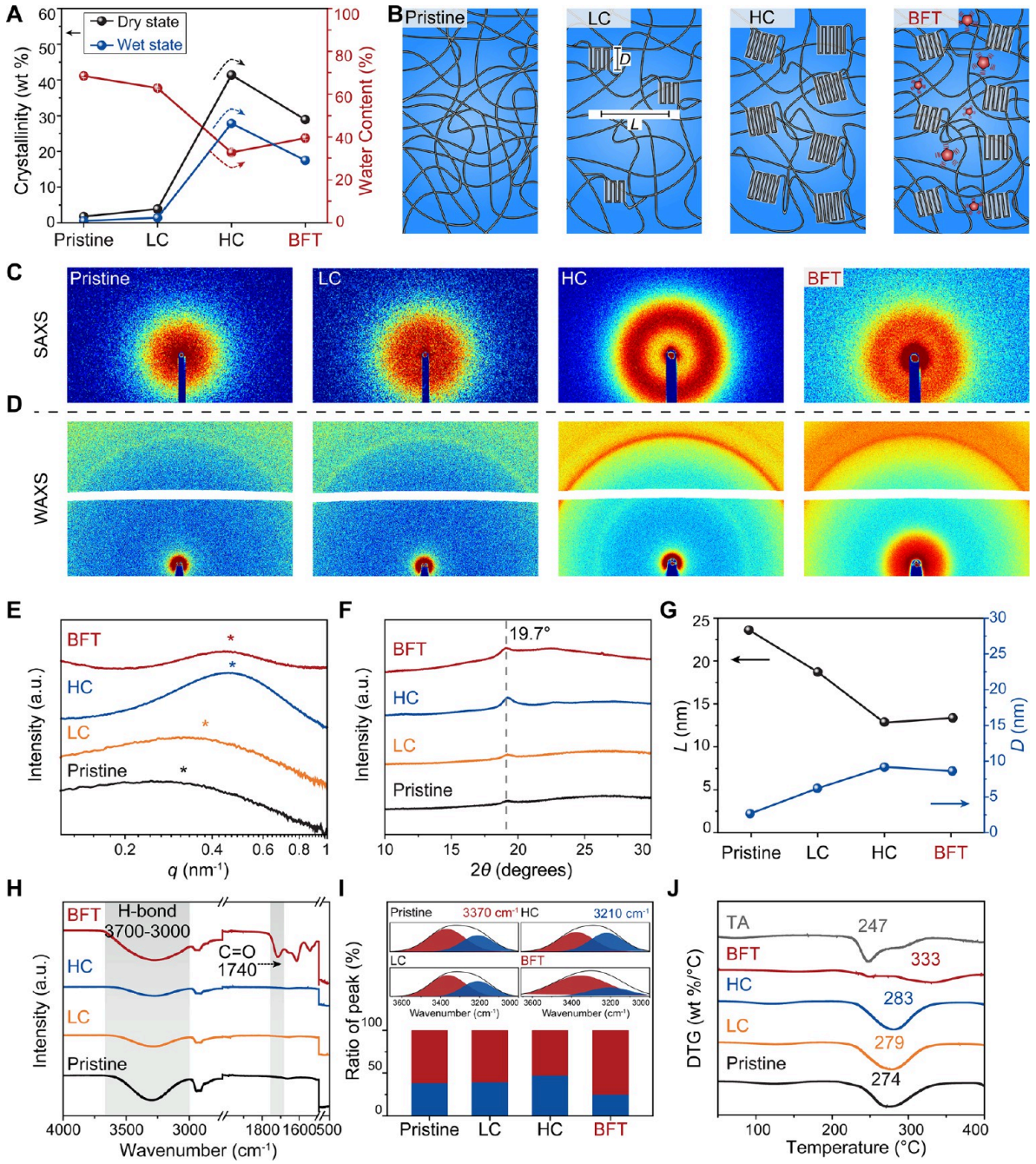


Figure 2. Characterization of hydrogel patches and their structural evolution. (A) Crystallinity and water content comparison of pristine, LC, HC, and BFT patches in dry and wet states. (B) Schematic illustrations showing the microstructural differences between pristine, LC, HC, and BFT patches, highlighting the formation of nanocrystalline domains and TA-mediated hydrogen bonds in BFT patches. (C) Small-angle Xray scattering (SAXS) patterns showing the nanostructure of different patches. (D) Wide-angle X-ray scattering (WAXS) patterns demonstrating the crystalline structures in various patches. (E) SAXS intensity profiles indicating the structural changes in different patches. (F) WAXS intensity profiles showing the diffraction peak at 19.7° associated with crystalline domains. (G) Distance between crystalline domains (L) and Crystalline domain size (D) calculated from SAXS and WAXS data. (H) ATR-FTIR spectra displaying the hydrogen bond and C=O stretching regions, illustrating the differences in chemical bonding among the patches. (I) Ratio of FTIR peak intensities highlighting the differences in hydrogen bond strength and content across the patches. (J) Differential thermogravimetric (DTG) analysis showing the thermal stability and decomposition temperatures of different patches, with BFT patches exhibiting the highest thermal stability due to dual cross-linking.

facilitate crystallization. In moist conditions, the hydroxyl groups in PVA's amorphous regions form enhanced hydrogen bonds, particularly when aided by hydrogen bond enhancers like polyphenols. These properties allow PVA to mimic spider silk multiscale molecular architecture. The highly ordered amino acid sequences in spider silk form stable β-sheet structures, which organize into nanoscale crystalline domains. By thermally annealing dry-cast PVA films under pressure, we induced PVA molecular chains to rearrange and form abundant nanocrystalline domains. Despite crystallization, PVA retains amorphous regions, enabling hydrogen bonding between chains, which is further enhanced in the presence of tannic

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acid (TA), a polyphenol used to noncovalently cross-link the crystallized PVA films (Figure S1A). After thorough washing, the resulting BFT hydrogel patch remains soft and suturable, allowing secure attachment to tissues through suturing (Figure 1F). Its postsuturing robustness is demonstrated by its ability to lift a 1 kg weight using medical 4−0 sutures (Figure 1G). Polarized light analysis during tensile testing reveals that this biomimetic BFT patch exhibits enhanced crack blunting and significant molecular chain orientation in the crack propagation zone, outperforming the pristine dry-cast film (Figure 1H).

In this study, we categorized PVA patches into four groups to investigate the influence of different microstructures on multiscale stress deconcentration and mechanical performance: the pristine (dry-cast) patch, low-cross-linking (LC) patch (low-temperature annealing), high-cross-linking (HC) patch (high-temperature annealing), and the BFT patch (high-temperature annealing followed by TA cross-linking) (Figure S1B). The pressure annealing process, performed under controlled pressure, ensures uniform heating and annealing of the PVA film, which promotes nanocrystalline domain formation and growth. Annealing enhances the mobility and alignment of PVA molecular chains, with higher temperatures resulting in increased crystallinity due to the growth of nanocrystalline domains. This crystallization process reduces the hydrophilicity of PVA chains, leading to lower equilibrium water content in the swollen patch, and thus the crystallinity increases with higher annealing temperatures in both dry and wet states (Figure 2A, Figure S2A).

However, after TA cross-linking, the crystallinity and water content trends reverse. We propose that TA, when introduced to crystallized PVA, enhances hydrogen bonding in the amorphous regions and competes with intramolecular interactions within nanocrystalline domains (Figure 2B). This competitive interaction pulls some PVA chains from the crystalline into the amorphous regions, increasing hydrophilicity and water content. Consequently, the incorporation of TA increases the cross-linking density and hydrophilicity of the PVA patch. Microscopy reveals that the transition from pristine to BFT patch results in a gradual decrease in pore size, making the material denser (Figure S3).

Subsequent small-angle X-ray scattering (SAXS) and wide-angle X-ray scattering (WAXS) analyses revealed microstructural changes in PVA induced by different processing methods (Figure 2C and D). The 2D SAXS mapping shows that as annealing temperature increases, scattering intensity also increases, consistent with the crystallinity results, though it decreases slightly after TA cross-linking. The 1D plots derived from SAXS reveal the scattering intensity versus scattering vector (q), and the peak positions of q were used to calculate the average spacing of crystalline domain centers (L value) according to Bragg's law (Figure 2E and G). The results show a progressive increase in the L value from the pristine, LC, to HC groups, indicating that pressure annealing enhances nanocrystalline density. However, TA treatment leads to an increase in the L value compared to the HC group, suggesting sparser crystalline domains. From the perspective of amorphous−crystalline interactions, TA affects both phases by cross-linking amorphous chains, restricting mobility, and disrupting crystalline order, increasing the amorphous fraction. These effects collectively alter chain mobility and network relaxation, making the increase in intercrystalline distance (L value) a result of both crystalline disruption and amorphous phase reorganization. The average size of crystalline domains (D value), calculated from the full width at half-maximum of the characteristic diffraction peaks in WAXS, also increases with annealing temperature but reverses after TA treatment (Figure 2F and G). These findings indicate that pressure annealing significantly enhances PVA molecular chain interactions, resulting in denser and larger nanocrystalline domains. However, TA cross-linking disrupts some of these domains, an effect consistent with crystallinity trends shown in Figure 2A.

To examine changes in intermolecular interactions, we conducted infrared spectroscopy. The ATR-FTIR (attenuated total reflectance Fourier transform infrared spectroscopy) profile showed the characteristic C=O peak at 1714 cm^{-1}, indicating the formation of ester bonds and confirming the incorporation of TA (Figure 2H). To distinguish between strong and weak hydrogen bonds within the PVA hydrogel patch, we performed peak fitting on the spectrum (Figure 2I). Stronger hydrogen bonds exhibit higher bond energy and longer bond lengths, causing a red shift in the absorption peak, which reflects changes in hydrogen bonding strength. The peak at 3370 cm^{-1} corresponds to the stretching vibration of hydroxyl groups involved in strong hydrogen bonding (e.g., PVA-TA hydrogen bonds), while the peak at 3210 cm^{-1} is associated with weaker hydrogen bonds (e.g., PVA−PVA hydrogen bonds). Peak area ratio analysis indicates that, compared to the weaker hydrogen bond interactions between the −OH groups of PVA molecules, TA can form stronger hydrogen bonds with PVA,[29] further optimizing the molecular chain network structure of the patch. These processing methods regulate the hydrogen bond proportions, enhancing overall intermolecular interactions within the patch.

Derivative thermogravimetric (DTG) analysis further demonstrated increased thermal stability, progressing from the pristine to the HC patch (Figure 2J and Figure S2B). TA incorporation notably enhances thermal stability beyond the levels achieved by annealing alone, as TA introduces additional hydrogen bonding and cross-linking within the PVA matrix, restricting polymer chain mobility and raising the energy needed for decomposition. Additionally, we further confirmed the optimization of intermolecular interactions in the BFT patch induced by TA using a hydrogen bond disruptor (urea). HC and BFT patches were immersed in a 2 wt % urea solution (denoted as HC-U and BFT-U, respectively), and tensile tests were subsequently performed. The results indicated that the reductions in strength, modulus, and fracture energy for BFT-U were greater than those for HC-U (Figure S4). This phenomenon is attributed to the stronger hydrogen bonding interactions in the BFT patch after TA cross-linking, which leads to a more significant disruption of hydrogen bonding interactions in BFT-U upon urea treatment. These findings highlight how pressure annealing and TA cross-linking refine the multiscale structure of the PVA patch, effectively aligning with the spider silk-inspired design strategy and achieving a synergistic effect between nanocrystalline domains and intermolecular interactions.

Flaw Tolerance under Single Large-Range Deformation. In patches with high flaw tolerance, structural robustness is maintained even with the presence of defects, making them reliable for practical applications. For medical patches, effective stress deconcentration around flaws is essential to prevent rapid crack propagation. In PVA patches, the amorphous chains and crystalline domains orient under stress, altering birefringence, which allows in situ polarized light observation

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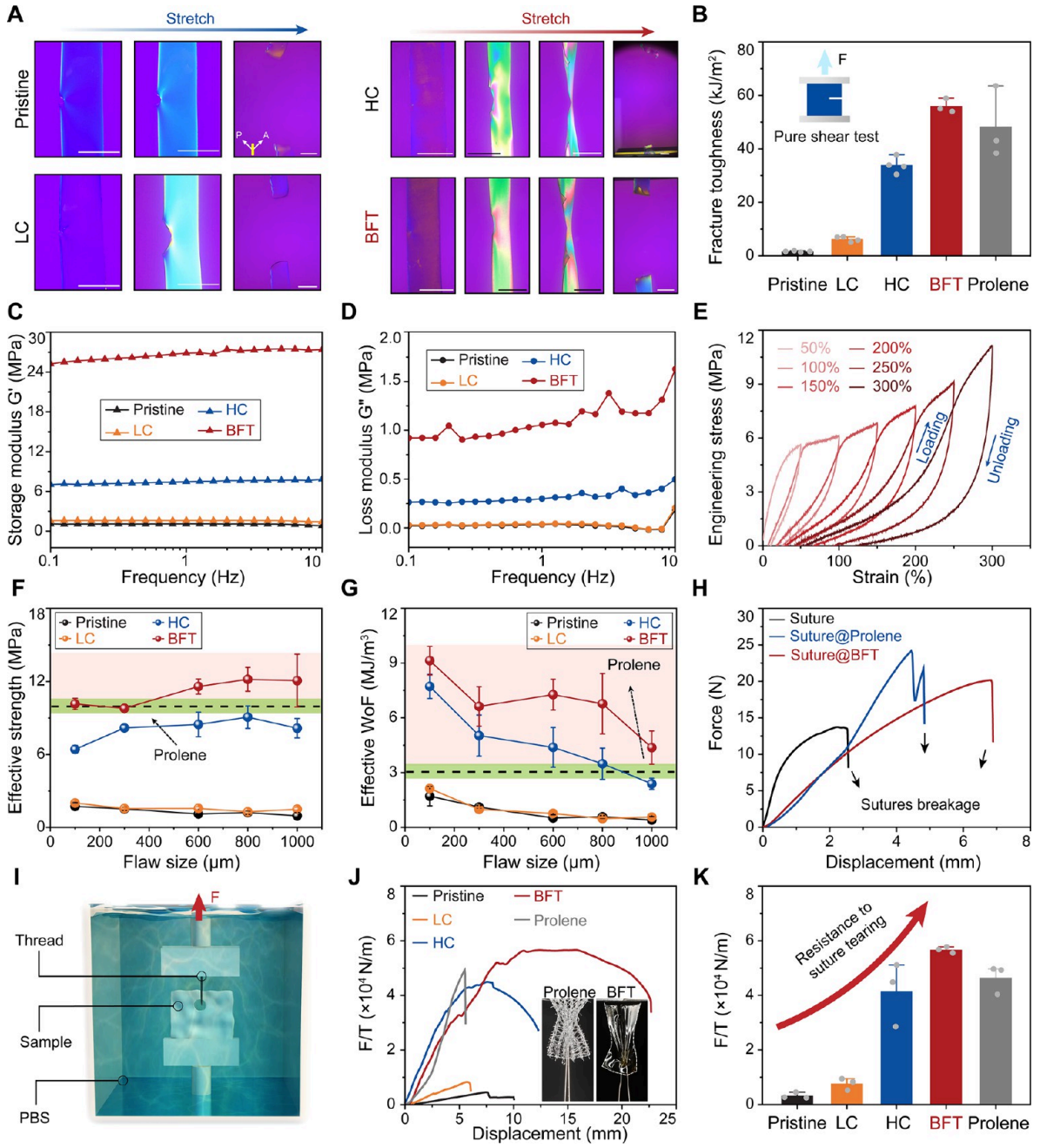


Figure 3. Mechanical properties and flaw tolerance of hydrogel patches. (A) Representative polarized optical images showing the birefringence of pristine, LC, HC, and BFT patches under uniaxial stretching. Three independent experiments were conducted with similar results. Scale bars: 5 mm. (B) Fracture toughness of pristine, LC, HC, BFT, and Prolene patches, measured by pure shear tests, demonstrating the superior toughness of BFT patches. (C–D) Dynamic mechanical analysis (DMA) showing the storage modulus (G') and loss modulus (G") of pristine, LC, HC, and BFT patches as a function of frequency, highlighting the mechanical stability of BFT patches under different loading conditions. (E) Engineering stress–strain curves during cyclic loading and unloading. (F-G) Residual strength and work of fracture of patches with varying flaw sizes, comparing the flaw tolerance of BFT patches with other samples and the Prolene mesh patch. (H) Force–displacement curves for suture tear resistance tests, comparing BFT and Prolene patches in sutured configurations. (I) Schematic of the setup for suture tear resistance tests in a PBS environment, replicating physiological conditions. (J) Force/tear displacement curves of sutured patches. (K) Bar graph summarizing the suture tear resistance (F/T) of the different patches.

to analyze stress distribution in different regions (Figure 3A, and movie S1). A notch was introduced in all four hydrogel patch groups to initiate cracks. Pristine and low-cross-linking (LC) patches showed limited birefringence regions and sharp crack tips, leading to rapid fracture. In contrast, high-cross-linking (HC) and bioinspired flaw-tolerant (BFT) patches exhibited expanding birefringence regions, indicating wider stress distribution and observed crack blunting. The polarization colors also varied, with more green-blue indicating greater orientation and thickness, while red-yellow showed less orientation. HC patches displayed concentrated blue-green areas at the notch edges, while the BFT patches had smoother transitions and smaller blue-green regions, suggesting more effective stress dispersion around flaws.

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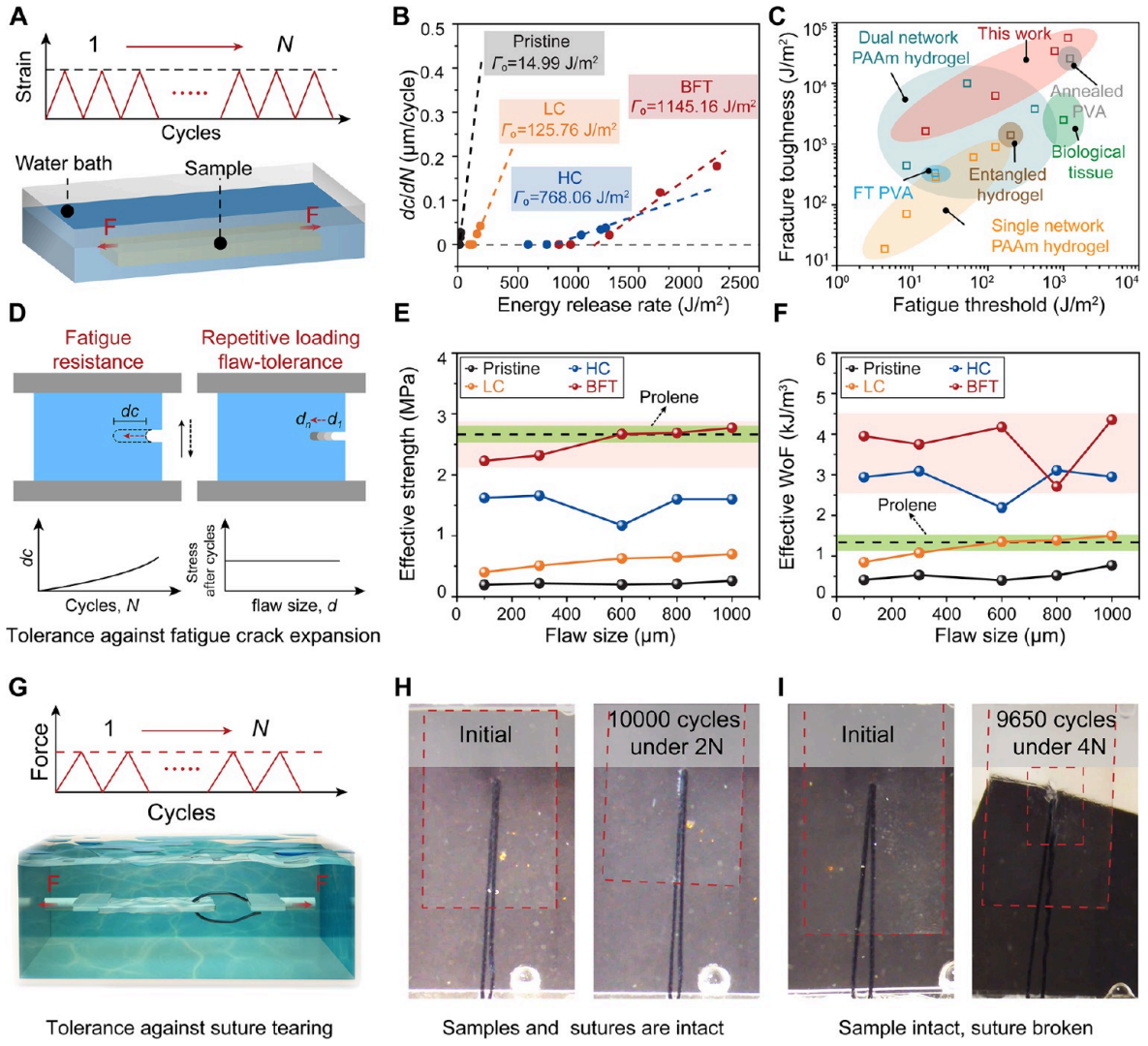


Figure 4. Fatigue resistance and flaw-tolerance of patches. (A) Schematic of cyclic loading test setup for evaluating fatigue threshold. (B) Plot of crack growth rate per cycle (dc/dN) as a function of energy release rate (G) for different patch groups: pristine, LC, HC, and BFT. (C) Ashby plot comparing fracture toughness and fatigue threshold among various hydrogel systems. (D) Illustration the difference between of fatigue resistance test and flaw-tolerance test under repetitive loading, emphasizing the tolerance against fatigue when samples with crack. (E-F) Residual strength and work of fracture versus flaw size for different patches after 5000 cycles under 50% strain, including a commercial Prolene patch. (G) Schematic of cyclic loading test setup for evaluating fatigue resistance to suture tearing. (H–I) Photographs showing the integrity of samples and sutures before and after cyclic loading under 2N and 4N loading.

According to the Griffith criterion and fracture mechanics, flaws lower the critical stress for fracture. To quantify crack resistance, fracture toughness was measured (Figure 3B), showing that an increase in nanocrystalline domains progressively enhanced toughness. In the BFT group, the synergy between nanocrystalline domains and intermolecular hydrogen bonding further improved toughness, comparable to commercial mesh patches, due to the enhanced energy storage and dissipation of the multiscale structure. Dynamic mechanical analysis (DMA) confirmed that nanocrystalline cross-linking and TA-reinforced hydrogen bonds increased the storage modulus, indicating rigidity (Figure 3C, D and Figure S5A). The BFT patch also showed a higher loss modulus, indicating enhanced energy dissipation, which was evident in hysteresis loop tests where the BFT group maintained a dissipation rate over 60% across strains (Figure 3E and Figure S6). Upon increasing temperature, the loss modulus of the BFT patch decreased to a level similar to HC, highlighting the influence of TA-mediated hydrogen bonds (Figure S5B). This approach effectively deconcentrates stress around flaws, improving resistance to flaw growth.

High flaw tolerance enhances durability, enabling patches to maintain performance despite defects. Patch flaw tolerance was evaluated using critical stress values across varying defect sizes (100–1000 μm) (Figure 3F). With a constant benchmark set by the engineering strength of porous Prolene (~9 MPa), pristine and LC patches displayed low strength, while HC patches, due to high nanocrystalline content, achieved tolerable strengths of 6–8 MPa but remained weaker than BFT patches. BFT patches, with optimized stress deconcentration, showed strength of 9–12 MPa in the presence of defects, comparable to Prolene. In terms of fracture work, both HC and BFT patches exceeded Prolene, enhancing robustness against deformation (Figure 3G).

When sutured with polypropylene 4–0 sutures, both Prolene and BFT patches delayed suture breakage, though

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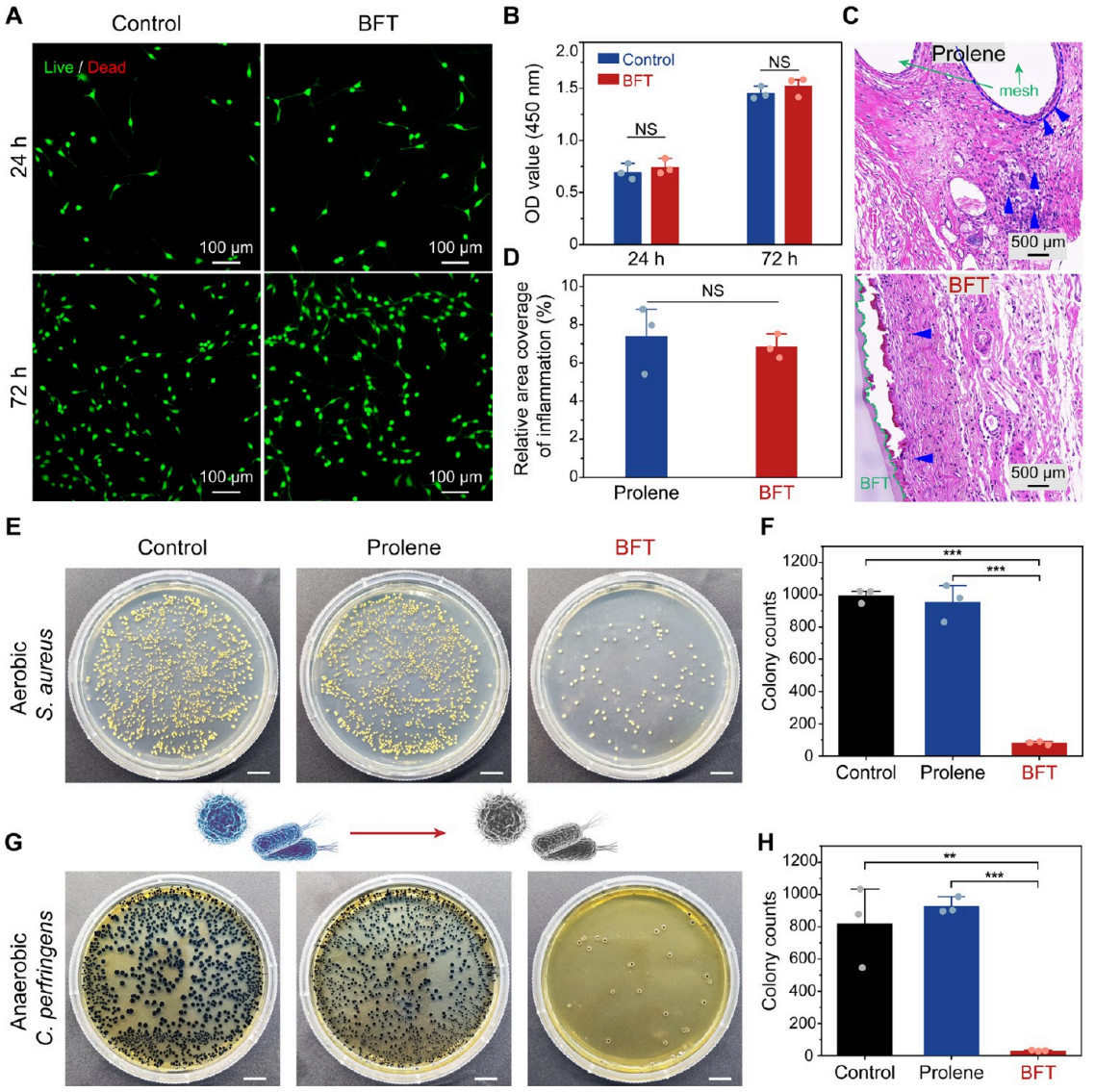


Figure 5. Biocompatibility and antimicrobial performance of BFT patches. (A) Representative images of LIVE/DEAD assay of fibroblasts cultured on control and BFT patches at 24 and 72 h. (B) CCK-8 assay results of fibroblast viability on Control and BFT patches at 24 and 72 h. (C–D) Representative H&E staining of tissue section images after 14 days of implantation (C), comparing inflammation between Prolene and BFT patches (D). Three independent experiments were conducted with similar results. (E-H) Representative digital photos of bacterial colony of aerobic S. aureus (E) and anaerobic C. perfringens (G) grown on control, Prolene, and BFT patches. Three independent experiments were conducted with similar results. Bacterial colony counts of aerobic S. aureus (F) and anaerobic C. perfringens (H) grown on control, Prolene, and BFT patches. Scale bars in (E) and (G) represent 10 mm. Values in (B), (D), (F), and (H), represent mean ± s.d. (n = 3 independent samples). Statistical significance and P values are determined by a two-sided Student's t test; NS: not significant, $^{*}P \leq 0.05$, $^{**}P \leq 0.01$, $^{***}P \leq 0.001$.

Prolene required higher force due to greater rigidity, while BFT displayed greater displacement, allowing for better energy absorption and reduced localized stress (Figure 3H and Figure S7). For tear resistance testing, we employed an experimental setup in PBS to simulate physiological conditions, comparing tear resistance per unit thickness (Figure 3I). The results demonstrated that BFT multiscale stress deconcentration provided superior tear resistance and displacement (Figure 3J and K). Overall, the BFT patch enhanced flaw tolerance and mechanical robustness make it a strong candidate for applications demanding durable and flexible sutured materials.

Flaw Tolerance under Cyclic Small-Range Deformation. Flaw tolerance under cyclic small-range deformation is critical for the durability and reliability of patch-based devices in practical applications, as the accumulation of fatigue damage over time can compromise their performance in demanding environments. To assess the fatigue resistance of patches under cyclic loading, it is necessary to determine the fatigue threshold ($G0$)—the energy release rate below which the material can withstand an infinite number of loading cycles without fatigue failure. To replicate real-world hydrated conditions, the tests were conducted in a water bath (Figure 4A). Samples with critical-sized defects were used to study fatigue behavior, as these defects serve as stress concentrators, essential for understanding crack initiation and propagation under cyclic loading. By applying cyclic loading at various strain levels and calculating the energy release rate (G) from the area under the steady-state stress–stretch curve of intact samples, we

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measured crack growth per cycle to gain insight into the material fatigue resistance.

As shown in Figure 4B, the fatigue threshold of the LC group (~125 J/m^2) is nearly ten times higher than the pristine group (15 J/m^2), demonstrating the critical role of nanocrystalline domains in resisting fatigue crack propagation. This effect is further amplified in the HC group ($G0$~770 J/m^2), where increased nanocrystalline domain content enhances fatigue resistance. Despite having lower crystallinity than the HC group, the BFT group achieved the highest fatigue threshold (~1150 J/m^2), indicating that enhanced hydrogen bonding can synergize with nanocrystalline domains to alleviate stress concentration at the crack tip (Figures S8–11). This multiscale stress deconcentration mechanism involves nanometer-sized crystalline domains dispersing local stress across a larger area, while TA-reinforced reversible hydrogen bonds in the polymer matrix absorb and dissipate cyclic stress energy, reducing crack tip stress intensity and enhancing fatigue resistance.

The Ashby plot comparing fracture toughness and fatigue thresholds of various hydrogels, including those in this study and from literature,[30−39] places the BFT hydrogel in the upper right corner, indicating both high fracture toughness and fatigue threshold (Figure 4C). This suggests that the BFT hydrogel not only resists crack initiation and propagation effectively but also maintains integrity under cyclic loading, outperforming many other hydrogels.

To further assess the robustness of patches under cyclic loading, retained effective strength and toughness were evaluated after 5,000 cycles at a strain level near the critical strain for crack propagation (50%) for each group (Figure 4D and Figure S12). The pristine and LC patches showed low retained strength across all flaw sizes, indicating poor cyclic loading resistance. HC patches, with superior crack pinning ability, retained higher strength (~1.5 MPa) after cycling. The BFT patch, with optimized multiscale stress dispersion and superior fatigue resistance, retained strength comparable to cycled Prolene across all flaw sizes (Figure 4E and Figures S13–17). Additionally, the BFT patch exhibited significantly higher fracture energy postcycling than Prolene, indicating enhanced defect fatigue tolerance (Figure 4F).

To simulate real-world durability, sutured patches were subjected to cyclic loading at different levels (2 and 4 N) in a water bath (Figure 4G-I). After 10,000 cycles at 2 N, both samples and sutures remained intact. After 9,650 cycles at 4 N, the sample remained intact while the suture broke, highlighting the superior flaw tolerance of the BFT hydrogel patch under high repetitive loading.

Biocompatibility and Antimicrobial Efficacy. By validating the biocompatibility of BFT patches, we aimed to demonstrate their feasibility as a replacement for mesh patches in medical applications. *In vitro* cytocompatibility was evaluated using live/dead staining and CCK-8 assays, with NIH-3T3 cells cocultured in BFT patch-conditioned medium. The results show that the number of cells increased over time in both control and BFT groups, with no significant dead cells observed (Figure 5A). Similar results were confirmed by the CCK-8 assay, showing no statistically significant differences between two groups (Figure 5B). Furthermore, we assessed the *in vivo* histocompatibility using hematoxylin-eosin (H&E) staining of tissue specimens after implanting Prolene and BFT hydrogel subcutaneously into rats for 14 days. No obvious inflammatory reaction was observed in the surrounding tissues (Figure 5C). Quantitative analysis of the relative area coverage revealed that the BFT hydrogel elicited a comparable inflammatory tissue response to the Prolene ($P > 0.05$, Figure 5D). These findings suggest that the BFT patch not only matches the mechanical performance of mesh patches, but also ensure a high level of biocompatibility when compared to Prolene patch, making it a promising alternative for medical applications.

The implantation process of patches is usually traumatic and prone to infection, especially in applications like abdominal wall defect repair, where postoperative aerobic or anaerobic bacterial infections can cause severe complications. The TA molecules used to enhance the weak hydrogen bonds in the BFT patch have been proven to achieve antibacterial effects through mechanisms such as metal ion chelation, which disrupts bacterial cell walls and affects enzyme activity. The antimicrobial efficacy of BFT patches was tested against aerobic *Staphylococcus aureus* (*S. aureus*) and anaerobic *Clostridium perfringens* (*C. perfringens*), representative of common pathogens in surgical site infections. The BFT patches were compared with blank control groups and Prolene patches to evaluate their relative antimicrobial effectiveness. The growth of *S. aureus* and *C. perfringens* on LB agar plates was significantly inhibited in the presence of BFT compared to Prolene (Figure 5E and G). Quantitative analysis revealed a significant reduction in CFUs compared to the control and Prolene groups (Figure 5F and H). To further explore the antimicrobial properties of the BFT patch, we examined the morphological changes in *S. aureus* and *C. perfringens* using SEM. As shown in Figure S18, after coculturing with the BFT patch, both *S. aureus* and *C. perfringens* exhibited shrunken, hollow, and engulfed morphologies, with some intracellular contents leaking from the bacteria, further supporting the antibacterial properties of the BFT patch. This antimicrobial property, combined with their superior mechanical performance and biocompatibility, makes BFT patches a promising alternative to conventional mesh patches for applications in medical surgeries, particularly in infection-prone environments like abdominal wall repair. The incorporation of TA not only improves the mechanical integrity of the patches but also provides an additional layer of protection against bacterial infections, ensuring safer and more effective surgical outcomes.

***In Vivo* Abdominal Wall Repair Evaluation.** In addition to the intrinsic antimicrobial properties provided by components such as tannic acid, PVA has been widely used in the development of antiadhesion barrier materials due to its excellent film-forming properties and biocompatibility.[40] Moreover, integral patches like the BFT patch could exhibit several performance advantages due to their continuous structure. These advantages stem from the absence of junctions and holes, which are inherent in mesh patches, leading to several key benefits in practical applications, particularly in medical and surgical scenarios. Integral patches lack the small crevices and corners present in mesh networks where contaminants can easily lodge and proliferate. Integral patches provide a uniform stress distribution across the entire patch, which ensures that the surrounding tissue experiences even loading, reducing the risk of localized tissue damage or failure. The continuous surface of integral patches promotes smooth integration with the host tissue, reducing the likelihood of adhesions that can compromise the repair process. The even surface and lack of fibers intersecting in different directions reduce the foreign body reaction, leading to a more

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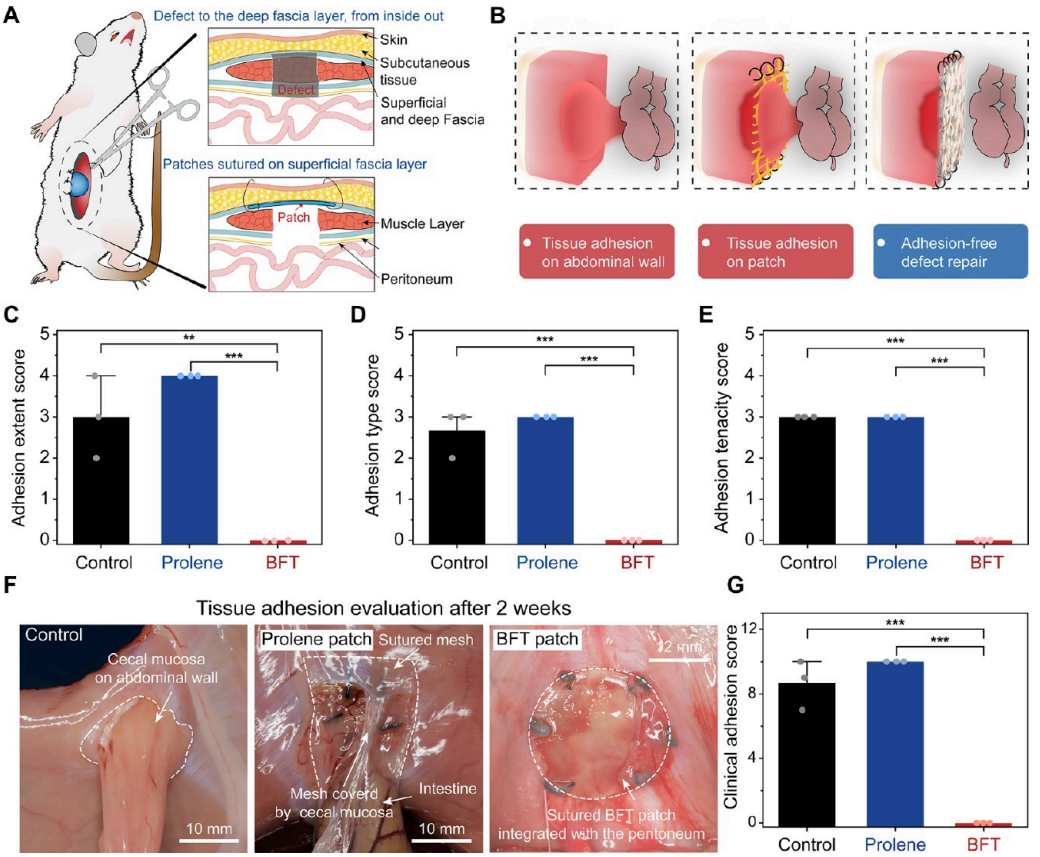


Figure 6. Evaluation of tissue adhesion in a rat abdominal defect repair model. (A) Schematic of the rat model used for abdominal defect repair, showing the defect location and the placement of patches on the superficial fascia layer. (B) Illustration of the different types of tissue adhesion observed in the study: tissue adhesion on the abdominal wall, tissue adhesion on the patch, and adhesion-free defect repair. (C) Quantification of the adhesion extent score for control, Prolene, and BFT patches after 2 weeks. (D) Quantification of the adhesion type score for the same groups. (E) Quantification of the adhesion tenacity score for the same groups. (F) Representative images showing tissue adhesion after 2 weeks in the control, Prolene, and BFT groups. Three independent experiments were conducted with similar results. (G) Quantification of the clinical adhesion score for the three groups. Scale bars in (F) represent 10 mm. Values in (C), (D), (E), and (G), represent mean ± s.d. (n = 3 independent samples). Statistical significance and P values are determined by a two-sided Student's t test; $^{**}P \leq 0.01$, $^{***}P \leq 0.001$.

biocompatible and less inflammatory response. To validate the practical advantages of the BFT integral patch over traditional Prolene mesh patches, a comparative study was conducted using a rat model for abdominal wall repair (Figure 6A). Abdominal wall patches, such as hernia patches, are subjected to unique mechanical challenges due to the dynamic nature of the human body. Actions such as running, bending, lifting, or twisting impose large stresses, while breathing, digestion, and subtle posture adjustments result in continuous, repetitive motions. All of these activities pose significant challenges to the suturing sites of the patch. Therefore, abdominal wall patch repair requires the patch to maintain robust support performance after suturing.

In this evaluation, we conducted a detailed assessment of tissue adhesion following abdominal wall repair in rats, comparing the outcomes between a control group (no patch repair), a group using commercial Prolene patches, and a group with BFT patches. The results showed that the control group exhibit higher adhesion scores across all parameters, indicating that in the absence of external intervention, tissue is prone to significant adhesion during the natural healing process, particularly between the cecal mucosa and the abdominal wall (Figure 6B). In contrast, the group using Prolene patches, although successfully covering the abdominal wall defect, experienced severe tissue adhesion due to the porous structure of the patch, resulting in the highest adhesion scores (Figure 6C, D, E, and G). Moreover, Figure 6F clearly shows that the Prolene patch adhered severely to the cecal mucosa, even causing partial intestinal obstruction. This phenomenon suggests that the mesh structure of the Prolene patch may serve as a breeding ground for tissue adhesion, leading to postoperative complications and repair failure. On the other hand, the BFT patch demonstrated a significant antiadhesion effect, with the lowest adhesion scores across all evaluated parameters (Figure 6C, D, E, and G). Figure 6F demonstrates that the BFT patch not only integrated effectively with the abdominal wall tissue but also exhibited minimal tissue adhesion. This indicates that the BFT patch, with its continuous hydrophilic polymer network structure, significantly reduces the tendency for tissue adhesion. Its smooth, nonporous surface effectively prevents tissue ingrowth, thereby enhancing the antiadhesion performance.

We further conducted a detailed evaluation of the biocompatibility and systemic toxicity of three different patch materials in the rat abdominal wall repair model. Through histological analysis of visceral tissue sections (Figure S19), we found that none of the major organs (such as the lungs, liver, spleen, kidneys, and heart) exhibited significant pathological

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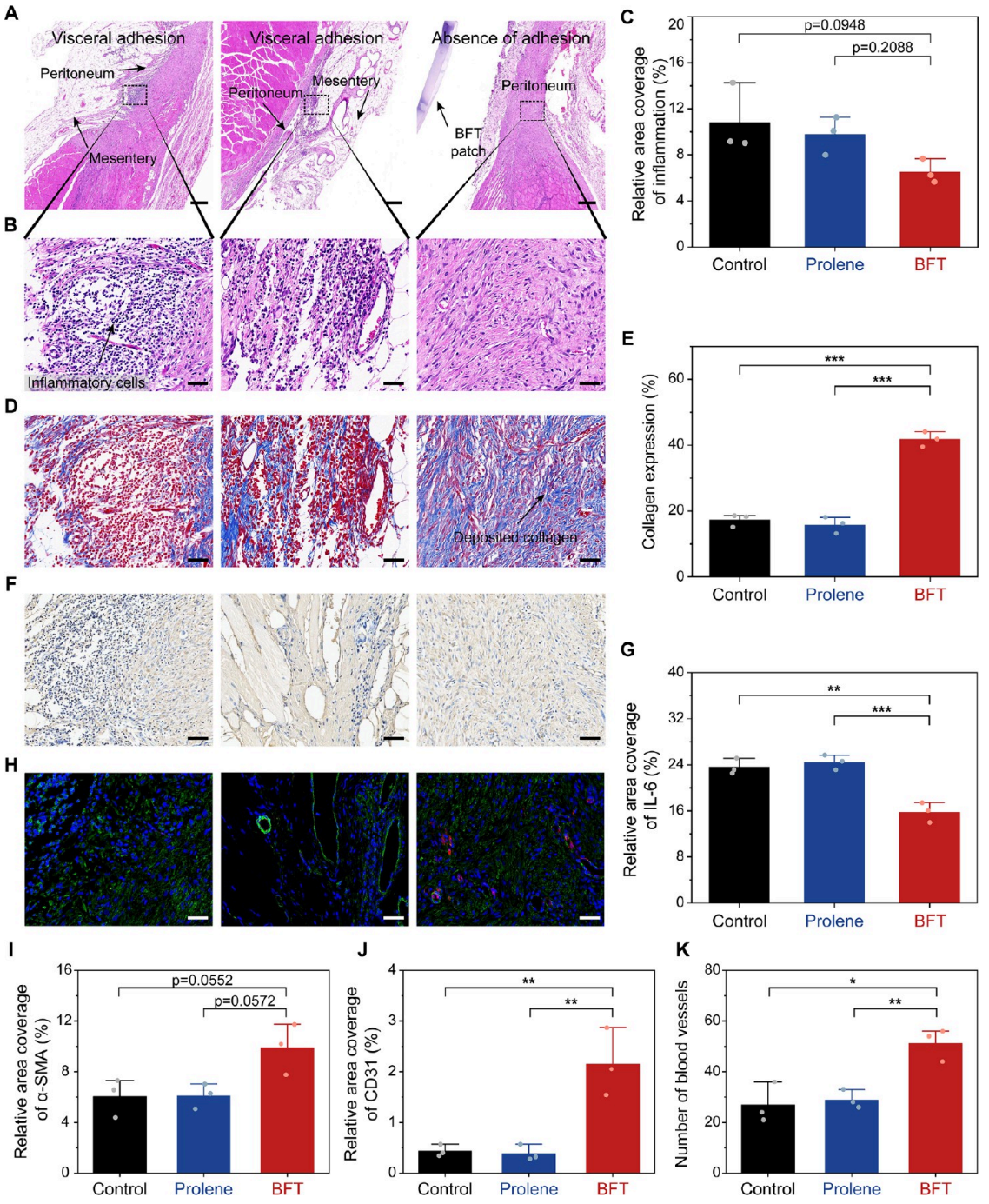


Figure 7. Histological analysis of tissue response after 14 days of implantation in a rat abdominal defect model. (A) Representative hematoxylin and eosin (H&E) stained sections of abdominal wall 2 weeks after repair, in the control, Prolene and BFT group. (B) Higher magnification images of the areas outlined in (A). (C) Quantification of the relative area coverage of inflammation across the three groups. (D) Masson's trichrome staining images for the tissues. For all the histology images, three independent experiments were conducted with similar results. (E) Quantification of collagen expression percentage across the three groups. (F) Immunohistochemical staining of IL-6 in abdominal wall tissues from the control, Prolene, and BFT groups, where the brown areas indicate IL-6 positivity. (G) Quantitative analysis of the relative area of IL-6-positive regions in the three groups. (H) Immunofluorescence staining of α-SMA (green) and CD31 (red) in abdominal wall tissues from the control, Prolene, and BFT groups. (I) Quantitative analysis of the relative area coverage of α-SMA fluorescence in the three groups. (J) Quantitative analysis of the relative area coverage of CD31 fluorescence in the three groups. (K) Quantification of new blood vessels in the wound tissue of the three groups based on fluorescence-stained tissue sections. Scale bars in (A) represent 500 μm; scale bars in (B), (D), (F) and (H) represent 50 μm. Values in (C), (E), (G), (I), (J) and (K), represent mean $\pm$ s.d. (n = 3 independent samples). Statistical significance and P values are determined by a two-sided Student's t test; NS: not significant ***$P \leq 0.001$.

changes in any of the groups, including the control, Prolene, and BFT groups. This indicates that the implantation of these patch materials did not induce noticeable visceral toxicity in the rats. In the analysis of the complete blood count tests, we

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observed that the BFT patch demonstrated excellent biocompatibility in rats, particularly in modulating inflammatory responses and maintaining normal red blood cell and platelet functions (Figure S20). Compared to the control group and the Prolene patches, the BFT patch significantly reduced white blood cell counts and the percentage of monocytes, indicating its advantage in minimizing overall immune response. Additionally, the BFT patch also showed regulatory effects on eosinophil and basophil levels, further suggesting its potential in alleviating inflammation and allergic reactions. The performance of hemoglobin and platelet-related indices indicated that the BFT patch maintained normal blood function comparable to the control group, and in some aspects, it outperformed the Prolene patches. Regarding blood biochemical parameters (Figure S21), we analyzed liver and kidney function-related markers (such as ALB, ALP, CK, CR, and TBL). The results showed no significant differences between the BFT group and the control or Prolene patches groups, indicating that the BFT patch performed well in maintaining liver and kidney function stability and did not cause muscle damage or myocardial stress responses.

To further evaluate the healing-promoting properties of the BFT hydrogel, histological analysis of the abdominal wall wounds was performed on the 14th postoperative day. From the H&E-stained tissue sections (Figure 7A and B), the blank group (without patch repair) exhibited a mild inflammatory response, with inflammatory cells primarily concentrated around the damaged area, although relatively evenly distributed. Due to the lack of mechanical support, the repair process may be insufficient, leading to lose tissue structure and severe visceral adhesion. In contrast, the Prolene patches group displayed a pronounced inflammatory response, particularly at the interface between the patch and the surrounding tissue. The porous structure of the Prolene patches facilitates the accumulation of inflammatory cells within its fibrous network, enhancing the local inflammatory response. Additionally, the mesh structure of the Prolene patches makes it more prone to visceral adhesion and infection, complicating the tissue repair process and limiting the regeneration of damaged tissue. These observations are partially reflected in the inflammation area statistics shown in Figure 7C, where, although there is no significant statistical difference, the distribution of inflammatory cells is more widespread and denser in the Prolene patches group according to histological tissue section observation. Conversely, the BFT patch group exhibited a distinctly different tissue response. Due to its integral structure, the BFT patch showed no significant accumulation of inflammatory cells in the repair area, with the inflammatory response confined to a smaller region. This suggests that the BFT patch can effectively suppress the inflammatory response, preventing excessive tissue irritation. More importantly, the BFT patch maintained its structural integrity throughout the repair process, effectively preventing visceral adhesions and significantly enhancing the quality of the repair. This structural stability was confirmed by a 14-day in vitro degradation study (Figure S22). Furthermore, the BFT patch exhibited excellent permeability (Figure S23), facilitating the effective transport of nutrients and gases during tissue healing, thereby providing an optimal environment for tissue regeneration. These advantages make the BFT patch particularly effective in abdominal wall defect repair. This conclusion is further supported by the results presented in Figures 7D and E, which show that collagen expression levels in the BFT patch group were significantly higher than in the other groups, highlighting its distinctive role in promoting tissue regeneration.

To further investigate the mechanisms by which BFT alleviates inflammation and promotes tissue repair, we tracked key biomarkers through immunohistochemical staining and immunofluorescence, including interleukin-6 (IL-6), α-smooth muscle actin (α-SMA), and CD31, to assess the inflammatory response and angiogenesis after tissue repair. First, the inflammatory levels of the tissues were quantitatively evaluated through IL-6 immunohistochemical staining. Compared to the control and Prolene groups, the expression of IL-6 in the BFT group was significantly reduced (Figure 7F and G), indicating that BFT effectively mitigates inflammation. On the other hand, angiogenesis is a key process in wound healing. Therefore, we analyzed the tissue's ability to form blood vessels using quantitative results for α-SMA (a marker of fibroblast), CD31 (a marker for endothelial cells of newly formed blood vessels), and the number of blood vessels. Compared to the control and Prolene groups, the wound sections in the BFT group exhibited higher expression of α-SMA, CD31, and greater blood vessel formation (Figure 7H–K), suggesting that BFT has a strong ability to promote angiogenesis and aid in wound healing.

CONCLUSIONS

This project successfully developed a bioinspired flaw-tolerant hydrogel patch, designed to address the limitations of conventional integral and mesh-based patches used in abdominal wall repair. By integrating multiscale stress deconcentration mechanisms, the BFT patch demonstrated significant advancements in both mechanical performance and biological compatibility. The BFT patch exhibited high retained strength and fracture energy, even in the presence of flaws at large-range loading and small-range cyclic loading. This suggests that the dual-cross-linking approach, which combines high-energy nanocrystalline domains with dynamic reversible bonds, effectively blunts cracks and deconcentrates stress, thereby enhancing robustness under large loading and durability under cyclic loading. The BFT patch also outperformed conventional Prolene patches in resistance to suture tearing, further highlighting its robustness in clinical scenarios that require high mechanical stability. Biocompatibility assessments showed that the BFT patch maintained cell viability comparable to the control group and induced more mild inflammation and tissue adhesion compared to Prolene patches.

The capability of BFT patch to combine mechanical toughness with biological antiadhesion addresses a critical need in surgical repair, particularly for applications that demand both high mechanical strength and minimal tissue adhesion, such as hernia repair. The dual-cross-linking strategy represents a significant innovation, allowing the patch to resist mechanical degradation from flaws introduced during surgical handling while minimizing adverse tissue interactions. The promising results of the BFT patch suggest potential applications beyond abdominal wall repair, including cardiovascular patches and wound dressings where both mechanical integrity and biocompatibility are paramount. Future work should focus on scaling up the production process, further optimizing the material properties for specific applications, and conducting long-term *in vivo* studies to fully establish the clinical efficacy and safety of the BFT patch. Additionally, integrating functional elements such as drug delivery systems

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or biosensors into the BFT hydrogel could open new avenues for multifunctional biomedical devices.

EXPERIMENTAL SECTION

Materials. PVA (M_w: 146000−186000; 99+% hydrolyzed) was purchased from Sigma-Aldrich (USA), and tannic acid (TA) (ACS reagent), glutaraldehyde (50 wt % in H_2O), urea (AR) and D-(+)-Glucose (AR) were purchased from Aladdin Industrial Corporation (Shanghai, China), and D-PBS (Dulbecco's Phosphate-Buffered Saline) buffer was purchased from Solarbio (Beijing, China). All reagents used in this study were used directly without further modification. Glucose test kit (S0201S) was purchased from Beyotime Biotechnology (Shanghai, China). Prolene polypropylene mesh (Prolene patches) was obtained from Johnson & Johnson (Shanghai) Medical Devices Co., and Mersilk sutures (silk braided nonabsorbable suture, 4−0) was purchased from Shanghai Ethicon Medical Technology Corporation.

Synthesis of PVA Hydrogels. The 10 wt % PVA solution was heated in a water bath at 97 °C with stirring for 8 h. Briefly, the solution was defoamed at 5000 rpm for 10 min, poured into a mold, and placed on a drying table at 40 °C for self-evaporation film formation. Then, the PVA films were further annealed for 30 min at certain temperatures (80 and 140 °C). PVA hydrogels (pristine, LC, and HC) were obtained by immersing unannealed and different temperature annealed PVA films in water to achieve an equilibrium swelling state. BFT hydrogel was obtained by immersing HC hydrogel in a 20 wt % TA solution for 48 h and soaking in water for 72 h to remove excess TA. HC and BFT were immersed in a 2 wt % urea solution for 2 h to obtain HC-U and BFT-U.

Tensile Tests. The tensile tests were performed on a universal mechanical testing machine (Instron 68SC-1 with a load cell maximum value of 500 N). The tests were performed in a room-temperature water bath to avoid water loss from the hydrogel during testing. In the tensile test, the samples were cut into standard dumbbell shapes 35 mm in length and 2 mm in width, and thickness measurements were taken using a digital thickness gauge, following ISO37. The testing speed was set at 100 mm min^{-1} for both the tensile and successive loading−unloading tests. The area under the stress−strain curve to failure was defined as the work of fracture.

Pure Shear Tests. A pure-shear test was conducted to measure the hydrogels' fracture toughness (Γ). In the test, the samples were cut into rectangles 20 mm wide and 4 mm long, with the notch length of the notched samples set between 1/4−1/2 of the sample width. The thickness of each specimen was measured with a digital thickness gauge. The testing speed was set at 100 mm min^{-1}. Unnotched samples were used to measure stress-tension curves and the initial clamp distance was recorded as L_0. The notched sample was used to measure the stretch at crack initiation, denoted as λ_c. The fracture toughness value (Γ) was obtained by multiplying the area under the stress−strain curve of the unnotched specimens at a stretch of λ_c by L_0, according to the following equation:

$$\Gamma = L_0 \int_0^{\lambda_c} S d\lambda$$

Where S is the measured stress.

Fatigue Tests. The single-notch method was performed to measure the hydrogels' fatigue resistance. The fatigue tests were performed on a universal mechanical testing machine (KZ-1100 with a load cell maximum value of 500 N, Kezhun, China). Fatigue testing was performed in a water bath to prevent dehydration of the hydrogel. Cyclic tensile tests were conducted using notched samples with an initial crack length (c_0) smaller than 1/5 of the width (w_0) of the sample. All stretch cycles were conducted continuously for 5000 repetitions to reach steady state (Figure S12). A digital camera (AF4915ZTL, Dino-Lite) was used to monitor the crack propagation of the hydrogel. The energy release rate (G) was obtained using

$$G(\lambda, N) = 2k(\lambda)c(\lambda)W(\lambda, N)$$

where k is a function of strain, determined to be $k = 3/\sqrt{\lambda}$, c is the crack length after the Nth cycle, and W is the strain energy density of an unnotched sample with the same dimensions and the same tensile strain after the Nth cycle.

Suture Tear Resistance Tests. The resistance to suture tearing was evaluated on a universal tensile machine (Instron 68SC-1 with a load cell maximum value of 500 N). The entire test was performed in a D-PBS solution. One side of the sample was secured with a fixture, while the other side was secured to another fixture using a 4−0 silk braided nonabsorbable suture, positioned 5 mm from the edge of the sample. The test was performed at a tensile speed of 100 mm min^{-1}. To account for variations in the thickness of the different samples, the force (F) obtained from the test was normalized by dividing it by the sample thickness, providing a quantification the ability of each sample to resist suture tearing.

Fatigue Resistance Test to Suture Tearing. The fatigue resistance test to suture tearing was performed on a universal tensile machine (KZ-1100 with a load cell maximum value of 500 N, Kezhun). The entire test was conducted in a D-PBS solution. One side of the sample was secured by a fixture, while other side was secured by a nonabsorbable 4−0 silk braided suture positioned 5 mm from the sample edge on another fixture. The sample and suture were subjected to cyclic loading by pulling the suture at different fixation force values (2 and 4 N), while observations were made to assess the fatigue resistance.

SEM Characterization. To characterize the effect of annealing and TA cross-linking on the hydrogel network structure, all hydrogel samples were immersed in deionized water for 24 h and then freeze-dried using a freeze-dryer (Scientz-12N/A). To characterize the morphological changes of bacteria after coculturing with the BFT patch, the bacteria were fixed in 4% paraformaldehyde, followed by supercritical dehydration. The dried hydrogel and bacteria samples were coated with a thin layer of gold using a gold sprayer, and the morphology and structure of the hydrogels were then imaged using a JEOL JSM-IT800 scanning electron microscope.

***In Situ* Tensile Polarization Detection.** The test was performed using a lab-built polarized light detection setup equipped with an in situ tensile stage. The samples were prepared as rectangular specimens with a width of 5 mm and a length of 40 mm, with a defect of approximately 500 μm cut on one side. The polarizer and analyzer were aligned at 90 deg to each other, both positioned at 45 deg to the longitudinal direction of the specimen. The sample was stretched at a speed of approximately 3 mm s^{-1}, and the birefringence phenomenon was detected in real time during the tensile process.

ATR-FTIR Characterization. ATR-FTIR spectra were acquired in the wavenumber range of 4000−400 cm^{-1} on a Thermo Fisher Nicolet FT-IR instrument equipped with an ATR accessory for the analysis of the chemical composition and intermolecular interactions of different PVA hydrogels. Using Origin software, the ATR-FTIR spectra were fitted within the range of 2980−3700 cm^{-1}, with peak positions fixed at 3210 and 3370 cm^{-1}. The Gaussian function was selected for the nonlinear fitting, and the region was sufficiently wide to allow the nonlinear least-squares method (Levenberg−Marquardt optimization) to re-evaluate the baseline. All Gaussian function fits exhibited high degrees of fit, with $R^2 > 0.97$. The area ratios of the peaks at 3210 cm^{-1} and 3370 cm^{-1} to the total peak area were calculated to quantify the hydrogen bonds in materials.

X-ray Scattering Characterization. The effect of annealing temperature and TA cross-linking on the morphology of PVA nanocrystals was investigated by small-angle X-ray scattering (SAXS) and wide-angle X-ray scattering (WAXS) measurements. These were carried out using SAXS Point 2.0 (Anton Paar, Austria) with an X-ray radiation wavelength of 0.15406 nm. Two-dimensional (2D) SAXS and WAXS patterns were acquired with an Eiger R 1 M X-ray detector with a resolution of 75 × 75 pixels, and the sample-to-detector distances for SAXS and WAXS tests were 576 mm and 112 mm, respectively. The 2D SAXS/WAXS images were further processed using FIT2D software. The average distance (L) between crystalline domains was calculated as

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$$L = \frac{2\pi}{q_{max}}$$

where q_{max} represents the critical vector corresponding to the peak intensity.

According to the Scherrer equation, the crystalline domain size (D) of PVA is calculated as

$$D = k\lambda/\beta\cos\theta$$

where k is a dimensionless shape factor that varies with the actual shape of the domains, λ is the diffraction wavelength of the X-rays, and θ is the Bragg angle.

Water Content Measurement. Hydrogel samples were immersed in deionized water for 7 days to achieve swelling equilibrium. The swollen hydrogel samples were then placed in an oven at 60 °C for 6 h, followed by transfer to an oven at 120 °C for 2 h to remove residual water. The weight of the PVA hydrogel before (m_w) and after (m_d) drying was measured using an analytical balance. The water content was calculated according to the following equation:

$$f_{water} = \frac{m_w - m_d}{m_w} \times 100\%$$

Crystallinity Content Measurement. The crystallinity of PVA hydrogels was measured using a DSC (TA, Q2000) instrument. The PVA hydrogels were first overcross-linked using glutaraldehyde to immobilize the amorphous molecular chains in the PVA hydrogels and minimize further formation of crystalline domains during air-drying. Briefly, the PVA hydrogel was immersed in 100 mL of deionized water, to which 5 mL of glutaraldehyde and 1 mL of hydrochloric acid were added, allowing for overcross-linking over a 1-h period. The overcross-linked PVA hydrogel was then washed and soaked in deionized water for 6 h to remove any excess glutaraldehyde. Afterward, the samples underwent a 3-h thermal treatment at 40 °C, followed by DSC testing.

For the DSC measurements, the total mass (m) of air-dried sample was measured, and the sample was placed in a Tzero pan. It was heated from 50 to 250 °C in a nitrogen atmosphere at a flow rate of 50 mL min^{-1} with a heating rate of 10 °C min^{-1}. The heat flow curve showed a narrow peak between 200 and 250 °C, corresponding to the melting of the PVA crystalline domains. The enthalpy of melting of the crystalline domains per unit mass of the dried sample was obtained by integrating the endothermic transition in this range.

Therefore, the mass of the crystalline domains ($m_{crystalline}$) can be calculated using the following equation:

$$m_{crystalline} = \frac{mH_{crystalline}}{H^0_{crystalline}}$$

where $H^0_{crystalline}$ = 138.6 J/g is the enthalpy of melting of 100 wt % crystalline PVA measured at the equilibrium melting point. Therefore, the crystallinity of PVA in the air-dried state, X_{dry}, is calculated by the following equation

$$X_{dry} = \frac{m_{crystalline}}{m}$$

The crystallinity of PVA at swelling equilibrium, $X_{swollen}$, can be calculated using the following equation

$$X_{swollen} = X_{dry}(1 - f_{water})$$

Thermogravimetric Characterization. We measured the thermal stability of patches using the TGA (Netzsch, TG209F1) instrument. For TGA measurements, the sample was raised from 30 to 800 °C at a rate of 10 °C min^{-1} with an air flow rate of 20 mL min^{-1}.

Dynamic Mechanical Measurement. The storage modulus (G′) and loss modulus (G″) of PVA hydrogels were tested using a Dynamic mechanical analyzer (DMA, TA Q850, USA). Temperature scanning measurements were performed in the range of 30 to 60 °C. Dynamic frequency sweep measurements were performed between 0.01 and 10 Hz. The tests were performed in tensile mode with an amplitude setting of 30 μm, and the entire process was carried out in an aqueous solution using an underwater device.

Antibacterial Performance Test. *Staphylococcus aureus* (*S. aureus*) and *Clostridium perfringens* (*C. perfringens*) were selected and cultured overnight in Luria−Bertani (LB) medium and Tryptose Sulfite Cycloserine Agar (TSC) medium, respectively. The *C. perfringens* cultures were maintained at 37 °C under anaerobic conditions using a Defendor AW500 system (Hua Yue, Guangzhou, China). Patch tablets (10 mm in diameter) were immersed in 1 mL diluted *S. aureus* or *C. perfringens* and incubated at 37 °C for 2 h. Subsequently, 100 μL *S. aureus* or *C. perfringens* culture solution was spread on the LB or TSC agar medium, respectively. After overnight incubation, the plates were observed, and colony counting was carried out.

Biocompatibility Evaluation. *In vitro*, biocompatibility was assessed using a cell counting kit (CCK-8) assay and live/dead test with mouse NIH/3T3 fibroblasts. Initially, BFT tablets (10 mm in diameter) were immersed in 10 mL Dulbecco's modified eagle medium (DMEM) containing 10% fetal bovine serum and 1% penicillin-streptomycin, and then incubated at 37 °C for 24 h to prepare the extracts. NIH/3T3 fibroblasts were then seeded at a density of 5×10^3 cells/well in 96-well plates and cultured overnight to allow cell adherence. The extracts were added to the wells after cell adhesion. At incubation for 1 day and 3 days, the supernatants were removed, and a 100 μL mixture containing 90 μL DMEM and 10 μL CCK-8 solution was added to each well. The plates were incubated at 37 °C for 2 h, and the optical density (OD) was measured at 450 nm.

For the live/dead assay, 1×10^5 cells were seeded into a 12-well plate and cultured overnight until adherence, and the extracts were added. On days 1 and 3 of culture, the supernatants were removed, and a working solution consisting of 10 μL calcein acetoxymethyl ester (Calcein-AM) and 15 μL propidium iodide (PI) in 5 mL phosphate-buffered saline (PBS) was added to each well. Subsequently, the plates were incubated at 37 °C for 15 min and then observed using a fluorescence microscope. Live cells exhibited green fluorescence and dead cells exhibited red fluorescence.

***In Vivo* Biocompatibility.** Animal studies were in full compliance with the guidelines of the Institutional Animal Care and Use Committee of Hangzhou Institute of Medicine (HIM), Chinese Academy of Sciences, China (no.AP 2024−07−0072). Six male Sprague−Dawley (SD) rats, weighing approximately 300 g, were used in the experiment. After anesthesia with isoflurane (1.5 ∼ 2.5% isoflurane in oxygen), longitudinal incisions were made on the back of each rat. Subsequently, three BFT hydrogels (5 mm in diameter) and PPM were carefully placed subcutaneously, respectively. At 7 and 14 days postoperation, species were collected and fixed with 4% paraformaldehyde. Then 6-μm-thick sections were prepared and stained with hematoxylin-eosin (H&E).

***In Vitro* Degradation Test of the BFT Patch.** The BFT patch was soaked in PBS solution for 24 h to reach swelling equilibrium. The swollen BFT patch was then immersed in PBS solution at 37 °C (pH = 7.4). At predetermined time points, the BFT patch was removed, rinsed with deionized water, and freeze-dried for 2 days. The remaining mass ratio of the BFT patch was calculated using the following formula:

$$\text{Remaining mass ratio}(\%) = \frac{W_t}{W_0} \times 100\%$$

Where W_0 and W_t respectively represent the dry weight of the initial hydrogel and the dry weight of the remaining hydrogel after degradation at different time points.

BFT Patch Permeability Test. The glucose delivery capacity of the BFT was tested using a Franz diffusion cell apparatus. At 37 °C, the liquid in the lower diffusion chamber was sampled every 24 h, and the glucose concentration of the lower solution was measured using a glucose detection kit. A glass bottle containing 10 mL of deionized water was sealed with a lid that had a 1 cm diameter circular hole. The Prolene, BFT, and PET were placed over the hole and sealed in place.

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These sample groups underwent a permeability test in a climate-controlled chamber maintained at 37 °C and 60% relative humidity. The mass of water inside the bottle was recorded at predetermined time points. Gas permeability was then calculated using the following formula:

$$\text{Gas permeability} = \frac{W_1 - W_2}{s \times t}$$

where W_1 is the initial mass of water, W_2 is the mass of water after different permeation times, s is the area of the hole on the bottle cap, and t is the permeation time.

Establishment and *In Situ* Repair of the Abdominal Wall Defect in Rats. Rats with an initial average weight of 300 g were anesthetized using isoflurane (1.5 ∼ 2.5% isoflurane in oxygen). Under aseptic surgical conditions, a round defect was created in the abdominal wall using a 10 mm standard biopsy perforator. The full thickness of the abdominal wall muscle was removed, while the peritoneum and skin were left intact. Subsequently, the defect was repaired using polypropylene mesh and BFT hydrogel, with fixation achieved through 4−0 nylon sutures. Rats without any implanted materials served as the blank group. Postoperatively, penicillin was administered intramuscularly daily for 3 days. After 2 weeks, all rats were euthanized, and their abdominal cavities were opened to expose the repair sites. The repair results were then scored from adhesion extent score, adhesion type score, and adhesion tenacity score (Table S1). Following fixation with 4% paraformaldehyde, H&E staining, Masson staining, immunohistochemistry for IL-6, and immunofluorescence for CD31 and α-SMA were performed. Quantitative analysis of sections was conducted using ImageJ software.

Statistical Analysis. Statistical analysis was performed using GraphPad Prism 9.5.1 software. Continuous variables were expressed as mean ± standard deviation (SD). Comparisons between two groups were made using Student's t test, while one-way analysis of variance (ANOVA) was employed for comparisons among multiple groups. Statistical significance was denoted as follows: NS (no significance), * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.4c17122.

Fabrication and structural evolution (Figure S1); thermal analysis (Figure S2); SEM images of patch structures (Figure S3); mechanical properties comparison before/after urea treatment (Figure S4); temperature-dependent dynamic mechanical analysis (Figure S5); cyclic loading analysis (Figure S6); suture integrity evaluation (Figure S7); effective remained strength of pristine (Figure S8), LC (Figure S9), HC (Figure S10), and BFT (Figure S11) patches with various flaw sizes; fatigue threshold measurement (Figure S12); residual strength after 5000 cycles for pristine (Figure S13), LC (Figure S14), HC (Figure S15), BFT (Figure S16), and Prolene (Figure S17) samples; bacterial morphology (Figure S18); histopathological analysis (Figure S19); blood count (Figure S20); blood biochemistry (Figure S21); adhesion scoring criteria (Table S1); mass change in PBS (Figure S22); nutrient delivery performance (Figure S23) (PDF)

In situ polarized light detection under tensile stress (Movie S1) (MP4)

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▲X.Z., X.N.L., and M.H.W. contributed equally to this work. Z.P., X.Z., and X.N.L. conceived the idea and designed the experiments. Z.M.X., W.F.C., J.J.Z., K.L., Y.S.Z., and M.H.W. designed and carried out the experiments of biocompatible and abdominal wall defect repair experiments. Z.P. and W.P.X. supervised the research. J.H. and P.Z. helped characterizations. X.Z. and X.N.L. performed the mechanical performance experiments and analyzed the data. X.Z., Z.P., M.H.W., and W.P.X cowrote the manuscript. All authors discussed the results.

Notes

The authors declare no competing financial interest.

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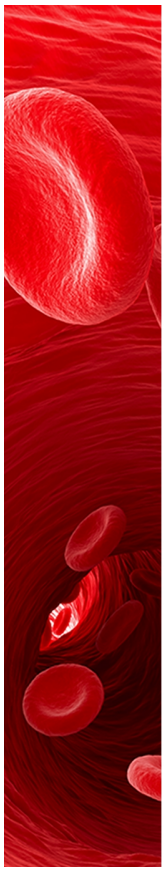
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